

Tools for Reactor-Antineutrino Studies in RAT-PAC

1.0.0

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Chapter 1

ROOT MACROS AND CLASSES

Instructions for using the (most useful) macros and classes in *github --> duvall3 --> rat-pac --> branch collab --> user --> root_macros*.

1.0.1 Installation

*Note: These are written for *nix systems (Mac/Linux)*

1. After git-cloning this repository (or downloading the individual files), run the `macro_paths.sh` script located in this directory and follow the instructions it provides.
2. If you are using the `rootlogon.C` provided in this directory, then everything should now be ready. If you are using your own `rootlogon.C` elsewhere, simply append the contents of this directory's version to yours.

1.0.2 Usage

Open ROOT and run `.x duvall3Include.cxx` to load the macros and classes.
For further information:

1. Start with the rest of this documentation;
2. If you don't find what you need, see the comments in the individual files; and finally,
3. If you still need assistance after that, feel free to contact me at *duvall3 dot git at gmail dot com*.

Chapter 2

Namespace Index

2.1 Namespace List

Here is a list of all documented namespaces with brief descriptions:

[TDuvalUtils](#)

Library of useful general-purpose functions for working with ROOT / RAT-PAC [9](#)

Chapter 3

Hierarchical Index

3.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

TClass	
TClassTemplate	17
TIBDParams	19
TRATGeo	20
TRATNeutronGen	22
TRATPACEventViewer	23
TRATVolume	25
TRefMatch	27

Chapter 4

Class Index

4.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

TClassTemplate	
Basic template for writing ROOT classes	17
TIBDParams	19
TRATGeo	20
TRATNeutronGen	22
TRATPACEEventViewer	23
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TRefMatch	
Implements KS/AD-test reference-matching algorithm	27

Chapter 5

Namespace Documentation

5.1 TDuvalUtils Namespace Reference

Library of useful general-purpose functions for working with ROOT / RAT-PAC.

Functions

- void [DumpHist](#) (TH1 *h)
Dump histogram to terminal.
- TEntryList * [EntryList](#) (const char *selection, TTree *T=0)
Shortcut for creating TEntryLists.
- void [ExportPlots](#) (const char *filename, const TString kGraphicsSaveFormat=".png")
Shortcut for printing all canvases in a file.
- TList * [FindMatchingObjects](#) (TCollection *colxn, TRegexp patternRE)
Scan a TCollection for an object whose name matches a regex.
- TList * [FindVarsOfType](#) (const char *varType="", Bool_t kCaseSensitive=kFALSE)
List global variables matching a specified type.
- TList * [ListFiles](#) (const char *pattern=".*.root")
Return a list of files matching a pattern.
- void [LoadAllKeys](#) ()
Shortcut to load all keys from the current file into memory ! USE WITH CAUTION !
- Double_t * [LogBins](#) (Double_t xmin=1.e-1, Double_t xmax=1.e2)
Return an array of logarithmically-spaced values.
- void [PrintBranches](#) (TObject *obj)
Print nicely-formatted summary of object's TBranches.
- Double_t [Prob2Sig](#) (Double_t prob)
Convert probability to significance.
- TH2D * [RadarPlot](#) (TH1D *h_in, Option_t *ho="cyllego", const Bool_t kNewCanvas=kTRUE)
Convert TH1D to radar plot.*
- void [ShiftStats](#) (TVirtualPad *p=gPad, Double_t deltaX=-.1, Double_t deltaY=0.)
Move the östats box out of the way.
- Double_t [Sig2Prob](#) (Double_t sig)
Convert significance to probability.
- Double_t [UnbinnedKSTest](#) (TTree *T1, TTree *T2, const char *branchName1, const char *branchName2="")
Apply unbinned KS test to a pair of TBranches.
- void [Zoom](#) (Double_t zoomFactor=2.)
Shortcut to set relative zoom.

5.1.1 Detailed Description

Library of useful general-purpose functions for working with ROOT / RAT-PAC.

Math:

- [EntryList\(\)](#)
- [LogBins\(\)](#)
- [Prob2Sig\(\)](#)
- [Sig2Prob\(\)](#)
- [UnbinnedKSTest\(\)](#)

Plotting:

- [DumpHist\(\)](#)
- [RadarPlot\(\)](#)
- [ShiftStats\(\)](#)
- [Zoom\(\)](#)

Finding Things:

- [FindMatchingObjects\(\)](#)
- [FindVarsOfType\(\)](#)
- [ListFiles\(\)](#)
- [PrintBranches\(\)](#)

File Operations:

- [ExportPlots\(\)](#)
- [ListFiles\(\)](#)
- [LoadAllKeys\(\)](#)

5.1.2 Function Documentation

5.1.2.1 DumpHist()

```
void T杜vallUtils::DumpHist (
    TH1 * h )
```

Dump histogram to terminal.

Dump histogram data as (x,y) pairs to stdout, along with a rough ASCII representation of the graph.

- Primarily for use in non-graphical sessions
- Note: Developed for use with *TH1D** and *TH1F** only
- Example:

```
TH1D* h = new TH1D("h", "h", 24, -3.5, 3.5);
h->FillRandom("gaus");
T杜vallUtils::DumpHist(h);
```

5.1.2.2 EntryList()

```
TEntryList * T杜vallUtils::EntryList (
    const char * selection,
    TTree * T = 0 )
```

Shortcut for creating TEntryLists.

Shortcut for creating a TEntryList from a TTree and a selection string.

Parameters

<i>selection</i>	– same as for TTree:Draw()
<i>T</i>	– pointer to the desired <i>TTree</i>

Return values

<i>eList</i>	– the resulting <i>TEntryList*</i>
--------------	------------------------------------

5.1.2.3 ExportPlots()

```
void T杜vallUtils::ExportPlots (
    const char * filename,
    const TString kGraphicsSaveFormat = ".png" )
```

Shortcut for printing all canvases in a file.

Extract all the TCanvas objects from a ROOT file and save them in the desired graphics format.

Parameters

<i>filename</i>	– desired ROOT file
<i>kGraphicsSaveFormat</i>	– .png, .eps, etc.

5.1.2.4 FindMatchingObjects()

```
TList * TDuvalUtils::FindMatchingObjects (
    TCollection * colxn,
    TRegexp patternRE )
```

Scan a TCollection for an object whose name matches a regex.

Scan a TCollection (*TList*, *TObjArray*, etc.) for an object whose name matches a regex.

ROOT's builtins can't search by PATTERN.

Note: It's hard to believe this isn't already a builtin function for all classes inheriting from TCollection; but if it exists, I haven't found it.

```
TObject* FindMatchingObjects( TRegexp regex )           // or
TObject* FindMatchingObjects( const char* pattern )
```

Parameters

<i>colxn</i>	– pointer to the desired <i>TCollection</i> object
<i>pattern(RE)</i>	– <i>char*</i> or <i>TRegexp</i> describing search pattern

Return values

<i>matchingObjs</i>	– <i>TList*</i> of matching objects
---------------------	-------------------------------------

5.1.2.5 FindVarsOfType()

```
TList * TDuvalUtils::FindVarsOfType (
    const char * varType = "",
    Bool_t kCaseSensitive = kFALSE )
```

List global variables matching a specified type.

List global variables whose type matches a specified pattern.

ROOT's builtins can't search by PATTERN.

Examples:

```
FindVarsOfType("canvas");           // or
TList *objarrList = FindVarsOfType("TObjArray");
```

Parameters

<i>varType</i>	– search string
<i>kCaseSensitive</i>	– whether search is case-sensitive

Return values

<i>oList</i>	– list of matching objects
--------------	----------------------------

5.1.2.6 ListFiles()

```
TList * T杜vallUtils::ListFiles (
    const char * pattern = ".*\.root" )
```

Return a list of files matching a pattern.

Return a TList of TSystemFiles in the current (system) directory whose names match a pattern.

5.1.2.7 LoadAllKeys()

```
void T杜vallUtils::LoadAllKeys ( )
```

Shortcut to load all keys from the current file into memory *! USE WITH CAUTION !*

Load all keys in current directory into memory

!!! USE WITH CAUTION !!!

Large files will overload memory and crash

5.1.2.8 LogBins()

```
Double_t * T杜vallUtils::LogBins (
    Double_t xmin = 1.e-1,
    Double_t xmax = 1.e2 )
```

Return an array of logarithmically-spaced values.

Wonderful shortcut for setting up log-binned histograms.

Lightly adapted from code graciously provided by Marc F. Bergevin.

Example:

```
Double_t *xBins = LogBins(1.e-1, 1.e5);
TH1D *h = new TH1D("h", "h", xBins);
h->FillRandom("landau");
h->Draw();
gPad->SetLogx(kTRUE);
```

5.1.2.9 PrintBranches()

```
void T杜vallUtils::PrintBranches (
    TObject * obj )
```

Print nicely-formatted summary of object's TBranches.

Print a TTree's branches in a format that is easier to scan visually than TTree::GetListOfBranches()->Is().

Parameters

<i>obj</i>	– <i>TTree*</i> or <i>TBranch*</i>
------------	------------------------------------

5.1.2.10 Prob2Sig()

```
Double_t TDuvalUtils::Prob2Sig (
    Double_t prob )
```

Convert probability to significance.

Simple function to convert a probability to a corresponding Gaussian significance level.
This probably already exists as a built-in function somewhere, but I'm adding it here for convenience.

Parameters

<i>prob</i>	– decimal probability on [0,1]
-------------	--------------------------------

Return values

<i>sigma</i>	– significance
--------------	----------------

5.1.2.11 RadarPlot()

```
TH2D * TDuvalUtils::RadarPlot (
    TH1D * h_in,
    Option_t * ho = "cyllego",
    const Bool_t kNewCanvas = kTRUE )
```

Convert TH1D* to radar plot.

Simple macro to redraw any 1-D histogram as a radar plot.

Parameters

<i>h_in</i>	– input histogram (<i>TH1D*</i>)
<i>ho</i>	– drawing option(s)
<i>kNewCanvas</i>	– when <i>kTRUE</i> , creates a new <i>TCanvas</i> rather than drawing over the current graphics pad

Return values

<i>h_out</i>	– new "radar" plot (<i>*TH2D*</i>)
--------------	--------------------------------------

5.1.2.12 ShiftStats()

```
void TDuvalUtils::ShiftStats (
    TVirtualPad * p = gPad,
    Double_t deltaX = -.1,
    Double_t deltaY = 0. )
```

Move the `stats` box out of the way.

Convenience function to translate the stats box.

Primarily intended for keeping the stats box from blocking the top of the color scale when histograms are drawn with "colz" and similar options.

5.1.2.13 Sig2Prob()

```
Double_t TDuvalUtils::Sig2Prob (
    Double_t sig )
```

Convert significance to probability.

Simple function to a Gaussian convert significance level to its corresponding probability.

This probably already exists as a built-in function somewhere, but I'm adding it here for convenience.

Parameters

<i>sig</i>	– significance
------------	----------------

Return values

<i>prob</i>	– decimal probability on [0,1]
-------------	--------------------------------

5.1.2.14 UnbinnedKSTest()

```
Double_t TDuvalUtils::UnbinnedKSTest (
    TTree * T1,
    TTree * T2,
    const char * branchName1,
    const char * branchName2 = "" )
```

Apply unbinned KS test to a pair of TBranches.

Function to execute *unbinned* Kolmogorov-Smirnov Test on a pair of TTrees.

Note: Branches must be of type Double_t.

See the notes in TMath::KolmogorovTest() and TH1::KolmogorovTest() for further details.

Parameters

<i>T1, T2</i>	– input trees
<i>branchName1</i>	– name of variable in <i>T1</i>
<i>branchName2</i>	– name of variable in <i>T2</i> (if different from <i>branchName1</i>)

Return values

<i>P</i>	– probability that the distributions were drawn from the same parent distribution, according to the K-S Test
----------	--

5.1.2.15 Zoom()

```
void TDuvallUtils::Zoom (
    Double_t zoomFactor = 2. )
```

Shortcut to set relative zoom.

Simple shortcut for adjusting zoom when running interactively.

Chapter 6

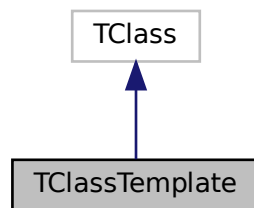
Class Documentation

6.1 TClassTemplate Class Reference

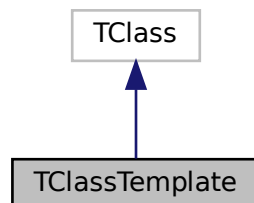
Basic template for writing ROOT classes.

```
#include <TClassTemplate.h>
```

Inheritance diagram for TClassTemplate:



Collaboration diagram for TClassTemplate:



Public Member Functions

- [TClassTemplate](#) ()
Default ctor.
- [TClassTemplate](#) (const char *someArg1, Double_t someArg2)
Normal ctor.
- Double_t [SomeCalculation](#) (Int_t someArg1=0)
Regular function with default argument.

Private Member Functions

- void [SomePrivateMethod](#) ()
Private method.

Private Attributes

- Double_t [fSomeData](#)
Data member.

6.1.1 Detailed Description

Basic template for writing ROOT classes.

6.1.2 Constructor & Destructor Documentation

6.1.2.1 TClassTemplate()

```
TClassTemplate::TClassTemplate (  
    const char * someArg1,  
    Double_t someArg2 )
```

Normal ctor.

Parameters

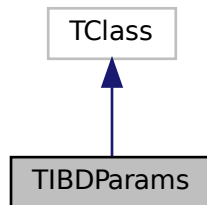
<i>someArg1</i>	– e.g., a filename
<i>someArg2</i>	– e.g., a track ID

The documentation for this class was generated from the following files:

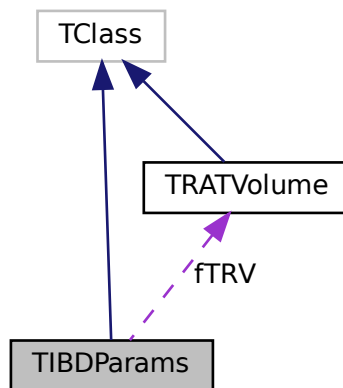
- TClassTemplate.h
- TClassTemplate.cxx

6.2 TIBDParams Class Reference

Inheritance diagram for TIBDParams:



Collaboration diagram for TIBDParams:



Public Member Functions

- **TIBDParams** ([TRATVolume](#) *vol)
- **TIBDParams** (Double_t reactorNuRate, Double_t standoff, Double_t nH, [TRATVolume](#) *TRV)
- Double_t **GetXS** ()
- Double_t **GetXSBarns** ()
- Double_t **GetThreshold** ()
- Double_t **GetFluxPowerRatio** ()
- Double_t **GetMuNeutronFlux** ()
- void **SetReactorPower** (Double_t reactorPower)
- void **SetReactorNuRate** (Double_t reactorNuRate)
- void **SetStandoff** (Double_t standoff)
- void **SetHydrogenDensity** (Double_t nH)

- void **SetVolume** ([TRATVolume](#) *TRV)
- Double_t **GetReactorPower** () const
- Double_t **GetReactorNuRate** () const
- Double_t **GetStandoff** () const
- Double_t **GetHydrogenDensity** () const
- [TRATVolume](#) * **GetTRATVolume** () const
- Double_t **NuFlux** ()
- Double_t **IBDVoiRate** ()
- Double_t **MuNeutronRate** ()
- Double_t **IBDRate** ()

Private Attributes

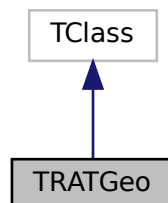
- Double_t **fReactorPower**
- Double_t **fReactorNuRate**
- Double_t **fStandoff**
- Double_t **fnH**
- [TRATVolume](#) * **fTRV**

The documentation for this class was generated from the following files:

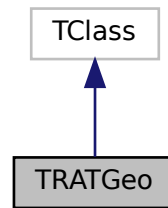
- TIBDParams.h
- TIBDParams.cxx

6.3 TRATGeo Class Reference

Inheritance diagram for TRATGeo:



Collaboration diagram for TRATGeo:



Public Member Functions

- void **Build** (const char *tcRegexp)
- void **Build** ()
- void **BuildCheckerboard** (const Int_t kDims=3)
- TFile * **GetFile** () const
- virtual const char * **GetFileName** () const
- TString **GetExperiment** () const
- TString **GetExperimentPath** () const
- TMap * **GetDB** () const
- TList * **GetListOfVolumes** () const
- [TRATVolume](#) * **GetVolume** (const char *volumeName)
- Int_t **GetEntries** () const
- TList * **GetCheckerboardActive** () const
- TList * **GetCheckerboardInert** () const
- TList * **FindVolumesContaining** (TVector3 location, Bool_t kPrint=kFALSE)
- TList * **FindVolumesContaining** (Double_t x, Double_t y, Double_t z, Bool_t kPrint=kFALSE)
- [TRATVolume](#) * **FindLowestVolumeContaining** (TVector3 location, Bool_t kPrint=kFALSE)
- [TRATVolume](#) * **FindLowestVolumeContaining** (Double_t x, Double_t y, Double_t z, Bool_t kPrint=kFALSE)
- void **ShowVolume** (const char *volumeName)
- void **ShowAll** ()
- void **Names** ()
- void **Types** ()
- void **Materials** ()
- void **Mothers** ()
- void **Sizes** ()
- void **Positions** ()

Private Member Functions

- void **Init** ()
- void **FindExperiment** ()
- void **FindCheckerboardActive** (Int_t kDims)
- void **FindCheckerboardInert** (Int_t kDims)

Private Attributes

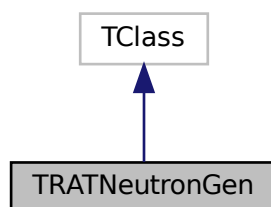
- TFile * **fFile**
- const char * **fFileName**
- TString **fExperiment**
- TString **fExperimentPath**
- const TMap * **fDB**
- TList * **fVolumeList**
- TList * **fActiveCells**
- TList * **fInertCells**

The documentation for this class was generated from the following files:

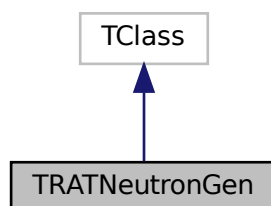
- TRATGeo.h
- TRATGeo.cxx

6.4 TRATNeutronGen Class Reference

Inheritance diagram for TRATNeutronGen:



Collaboration diagram for TRATNeutronGen:



Public Member Functions

- TVector3 * **KEtoMom** (Double_t KE)
- void **Generate** (Long64_t numNeutrons)
- Double_t **GetNeutronMass** () const
- TF1 * **GetSpectrum** () const
- TH1D * **GetSpecHist** () const
- TTree * **GetTree** () const
- char * **GetOutFilename** () const
- void **SetOutFile** (const char *newOutFile)
- void **DrawSpectrum** ()

Private Attributes

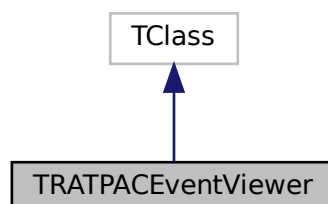
- Double_t **fNeutronMass**
- TF1 **fNeutronSpectrum**
- TH1D **fSpecHist**
- TString **fOutFilename**
- TTree **fTree**

The documentation for this class was generated from the following files:

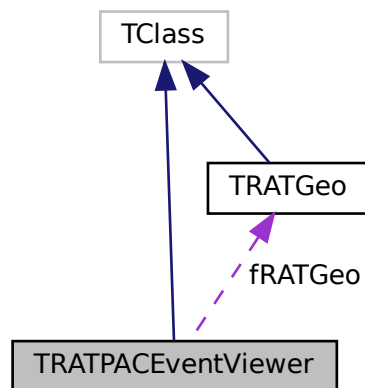
- TRATNeutronGen.h
- TRATNeutronGen.cxx

6.5 TRATPACEventViewer Class Reference

Inheritance diagram for TRATPACEventViewer:



Collaboration diagram for TRATPACEventViewer:



Public Member Functions

- **TRATPACEventViewer** (const char *fileName, const char *volumePattern=".*.")
- **Init** (const char *fileName, const char *volumePattern=".*.")
- TString **GetFileName** ()
- TFile * **GetDataFile** ()
- TTree * **GetTree** ()
- TString **GetVolumePattern** ()
- TRATGeo * **GetRATGeo** ()
- TGeoManager * **GetGeoManager** ()
- Bool_t **GetHighlightMode** ()
- TCanvas * **GetCanvas** ()
- Long64_t **GetTotalEvents** ()
- Long64_t **GetCurrentEvent** ()
- void **SetData** (Double_t someArg)
- void **SetVolumePattern** (const char *pattern)
- void **SetHighlightMode** (Bool_t bValue=kTRUE)
- void **DrawGeometry** ()
- void **DrawTracks** (Long64_t event=0)
- void **DrawNextEvent** ()
- void **DrawPrevEvent** ()
- void **RedrawCurrentEvent** ()
- void **Zoom** (Double_t zoomFactor=2)
- void **ToggleInertVis** ()

Private Member Functions

- void **SetCurrentEvent** (Long64_t eventID=0)
- void **HighlightCells** ()

Private Attributes

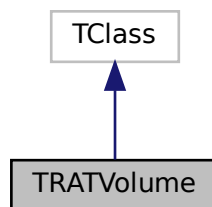
- TString **fFileName**
- TFile * **fDataFile**
- TTree * **fTree**
- TString **fVolumePattern**
- [TRATGeo](#) * **fRATGeo**
- TGeoManager * **fGeoManager**
- Bool_t **fkHighlight**
- TCanvas * **fCanvas**
- Long64_t **fTotalEvents**
- Long64_t **fCurrentEvent**

The documentation for this class was generated from the following files:

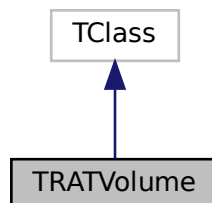
- TRATPACEEventViewer.h
- TRATPACEEventViewer.cxx

6.6 TRATVolume Class Reference

Inheritance diagram for TRATVolume:



Collaboration diagram for TRATVolume:



Public Member Functions

- **TRATVolume** (const char *volNameChr, const TMap *db)
- void **SetVolume** (const char *newNameChr)
- void **FindAbsolutePosition** ()
- virtual const char * **GetVolNameChr** () const
- const TString * **GetVolName** () const
- TFile * **GetFile** () const
- virtual const char * **GetFileName** () const
- TString **GetExperiment** () const
- TString **GetExperimentPath** () const
- TMap * **GetDB** () const
- TString **GetVolumeType** () const
- TString **GetMaterial** () const
- Double_t **GetDensity** () const
- TString **GetMother** () const
- TVector3 * **GetSize** () const
- Double_t **GetRMin** () const
- Double_t **GetRMax** () const
- Double_t **GetSizeZ** () const
- TVector3 * **GetRelativePosition** () const
- TVector3 * **GetAbsolutePosition** () const
- Double_t **Area** ()
- Double_t **AreaCM** ()
- Double_t **AreaMM** ()
- Double_t **Volume** ()
- Double_t **VolumeCM** ()
- Double_t **VolumeMM** ()
- Double_t **VolumeL** ()
- Double_t **Mass** ()
- Double_t **MassG** ()
- Double_t **MassTons** ()
- Double_t **NuFlux** (Double_t standoff, Double_t reactorNuRate)
- Double_t **IBDVoiRate** (Double_t standoff, Double_t reactorNuRate, Double_t nH)
- Double_t **IBDRate** (Double_t standoff, Double_t reactorNuRate, Double_t nH)
- void **PrintDerived** ()

Private Member Functions

- void **FindExperiment** ()
- void **FindVolumeType** ()
- void **FindMaterial** ()
- void **FindDensity** ()
- void **FindMother** ()
- void **FindRelativePosition** ()
- void **FindSize** ()
- void **FindAll** ()

Private Attributes

- const TString **defaultName** = "TRATVolume"
- const TString **defaultTitle** = "class for analyzing geometry from RAT-PAC ROOT file"
- const char * **defaultVolNameChr** = "volume_name"
- const TString **defaultVolName** = TString("volume_name")
- TString **keyStrVol**
- const char * **fVolNameChr**
- TString **fVolName**
- TFile * **fFile**
- const char * **fFileName**
- TString **fExperiment**
- TString **fExperimentPath**
- const TMap * **fDB**
- TString **fVolumeType**
- TString **fMaterial**
- Double_t **fDensity**
- TString **fMother**
- TVector3 **fSize**
- Double_t **fRMin**
- Double_t **fRMax**
- Double_t **fSizeZ**
- TVector3 **fRelativePosition**
- TVector3 **fAbsolutePosition**

The documentation for this class was generated from the following files:

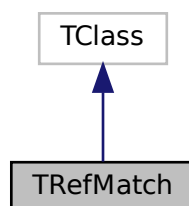
- TRATVolume.h
- TRATVolume.cxx

6.7 TRefMatch Class Reference

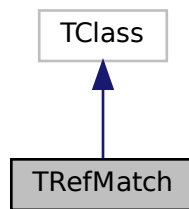
Implements KS/AD-test reference-matching algorithm.

```
#include <TRefMatch.h>
```

Inheritance diagram for TRefMatch:



Collaboration diagram for TRefMatch:



Public Member Functions

- [TRefMatch](#) ()
Default ctor.
- [TRefMatch](#) (const char *fileName, const char *treeName="T", const char *branchVarName="phi")
Regular ctor.
- void [Init](#) (const char *fileName, const char *treeName, const char *branchVarName)
Setter version of [Init\(\)](#)
- const char * [GetTestFileName](#) ()
- const char * [GetTestTreeName](#) ()
- const char * [GetTestVarName](#) ()
- TFile * [GetFile](#) ()
- TTree * [GetTree](#) ()
- TBranch * [GetBranch](#) ()
- const char * [GetReferenceTreeName](#) ()
- TList * [GetReferenceFileList](#) ()
- TFile * [GetOutFile](#) ()
- TRegexp [GetReferenceFilePattern](#) ()
- TSystemDirectory * [GetReferenceFileDir](#) ()
- Long64_t [GetnTestSampleEvents](#) ()
- Long64_t [GetnReferenceEvents](#) ()
- Bool_t [GetAnderson](#) ()
- Double_t [GetTestSamplePhiTrue](#) ()
- Double_t [GetProb](#) ()
- Double_t [GetSig](#) ()
- TMatrixD [GetResultsMatrix](#) ()
- TVectorD [GetResults](#) ()
- TFile * [GetBestMatchFile](#) ()
- TCanvas * [GetCanvas](#) ()
- TH1D * [GetTestSampleHist](#) ()
- TGraph * [GetResultsGraph](#) ()
- void [SetReferenceFileDir](#) (TSystemDirectory *refFileDir)
Set directory (by pointer) to search for reference files.
- void [SetReferenceFileDir](#) (const char *refFileDirName)
Set directory (by name) to search for reference files.
- void [SetReferenceFilePattern](#) (TRegexp patternRE)

- Set search pattern (by TRegexp object) for reference filenames.
 - void **SetReferenceFilePattern** (const char *pattern)
- Set search pattern (by string) for reference filenames.
 - void **SetReferenceFileList** (TList *fileList)
 - void **SetAnderson** (Bool_t bAnderson)
 - void **SetReferenceTreeName** (const char *treeName)
 - void **FillReferenceFileList** ()
- Populate the reference-file list based on the current reference-file directory and pattern.
 - void **SetnEvents** (Long64_t nTestSampleEvents=0, Long64_t nReferenceEvents=0)
 - Double_t **Prob2Sig** (Double_t prob)
- Convert probability to significance.
 - Double_t **Sig2Prob** (Double_t sig)
- Convert significance to probability.
 - void **ExtractRef** (const char *runName, const char *treeName="T", const char *branchName="phi")
- Extract distribution from reference run.
 - void **ExtractTest** (const char *runName, const char *treeName="T_ncap", const char *branchName="phi")
- Extract distribution from experimental run.
 - Double_t **UnbinnedKSTest** (TTree *T1, TTree *T2, const char *branchName1, const char *branchName2="", Long64_t nEvents1=0, Long64_t nEvents2=0)
- Apply unbinned Kolmogorov-Smirnov test.
 - Double_t **AndersonDarlingTest** (TTree *T1, TTree *T2, const char *branchName1, const char *branchName2="", Long64_t nEvents1=0, Long64_t nEvents2=0)
- Apply (unbinned) Anderson-Darling test.
 - void **RefCompare** ()
- Perform reference-comparison algorithm.
 - void **DrawResults** (Bool_t kDrawFit=kFALSE)
- Plot sample distribution and algorithm results.
 - void **PrintVerbose** ()
- Mostly settings.
 - void **PrintResults** ()
- Results summary.
 - void **SaveResults** ()
- Print canvas and save object.
 - void **Close** ()
- Close pads, files, etc.
 - void **Run** (Long64_t nTestSampleEvents=0)
- Execute a typical analysis.

Private Member Functions

- void **Init** ()
- Initialize: validate and fill members.
 - void **SetTestSamplePhiTrue** (Double_t phi)
 - void **SetProb** (Double_t prob)
 - void **SetSig** (Double_t sig)
 - void **SetResultsMatrix** (TMatrixD resmat)
 - void **SetResults** (TVectorD res)
 - void **SetBestMatchFile** (TFile *bestMatchFile)
 - void **SetCanvas** (TCanvas *c)
 - void **SetTestSampleHist** (TH1D *h)
 - void **SetResultsGraph** (TGraph *g)

Private Attributes

- const char * [fTestFileName](#)
Name of file containing data sample to be tested.
- const char * [fTestTreeName](#)
Name of tree containing data sample to be tested.
- const char * [fTestVarName](#)
Name of branch/variable containing data sample to be tested.
- TFile * [fTestSampleFile](#)
Address of test-sample file.
- TTree * [fTestSampleTree](#)
Address of test-sample tree.
- TBranch * [fTestSampleBranch](#)
Address of test-sample branch.
- const char * [fReferenceTreeName](#)
Name of tree in reference files.
- TList * [fReferenceFileList](#)
List of files containing reference distributions.
- TFile * [fOutFile](#)
Output file.
- TRegexp [fReferenceFilePattern](#)
Regex describing reference files.
- TSystemDirectory * [fReferenceFileDir](#)
Directory containing reference files.
- Long64_t [fnTestSampleEvents](#)
Number of events to use from test-sample tree (default value 0 will use all events)
- Long64_t [fnReferenceEvents](#)
Number of events to use from reference trees (default value 0 will use all events)
- Bool_t [fkAnderson](#)
If true, use Anderson-Darling rather than Kolmogorov-Smirnov.
- Double_t [fTestSamplePhiTrue](#)
The true value of phi for the test sample.
- Double_t [fProb](#)
Match probability (current entry)
- Double_t [fSig](#)
Match significance (current entry)
- TMatrixD [fResultsMatrix](#)
Matrix of comparison results.
- TVectorD [fResults](#)
3x1 vector containing best match value and corresponding probability and significance
- TFile * [fBestMatchFile](#)
Address of best-match reference file.
- Bool_t [fkHasInit](#)
Status indicator for whether Init has been called yet.
- Bool_t [fkHasRun](#)
Status indicator for whether RefComp has been called yet.
- TCanvas * [fCanvas](#)
Canvas for drawing results.
- TH1D * [fTestSampleHist](#)
TH1 for test-sample data.
- TGraph * [fResultsGraph](#)
TGraph for algorithm results.

6.7.1 Detailed Description

Implements KS/AD-test reference-matching algorithm.

Standard Usage ([R] = always required):

1. Instantiate (NOTE: must construct with "new") [R]
2. Call Init ("setter version") if used default ctor
3. Set reference directory / pattern if needed
4. Call FillReferenceFileList [R]
5. Set tree and branch names if needed
6. Set number of events to use from {test-sample, reference-distrib} if desired
7. Call RefCompare [R]
8. Call DrawResults if desired
9. Call Save if desired
10. Call Close when finished

6.7.2 Constructor & Destructor Documentation

6.7.2.1 TRefMatch()

```
TRefMatch::TRefMatch (
    const char * fileName,
    const char * treeName = "T",
    const char * branchVarName = "phi" )
```

Regular ctor.

Parameters

<i>fileName</i>	– name of file containing experimental/test data
<i>treeName</i>	– name of relevant TTree in input file
<i>branchVarName</i>	– shared variable name (b/t input and ref. files)

6.7.3 Member Function Documentation

6.7.3.1 AndersonDarlingTest()

```
Double_t TRefMatch::AndersonDarlingTest (
    TTree * T1,
```

```

TTree * T2,
const char * branchName1,
const char * branchName2 = "",
Long64_t nEvents1 = 0,
Long64_t nEvents2 = 0 )

```

Apply (unbinned) Anderson-Darling test.

Function to execute *unbinned* TMath::KolmogorovTest on a pair of TTrees containing TBranches with matching names – Usage: Double_t P = AndersonDarlingTest(TTree T1, TTree *T2, const char branchName) – Branches must be of type Double_t – P is the probability for match – *T1 and *T2 are pointers to the two input trees – branchName{1,2} are the branch/variable names in the respective trees – nEvents{1,2} are the number of entries to use from each tree (default value 0 will use all entries) – See the notes in ROOT::Math::GoFTest for details

6.7.3.2 Init()

```

void TRefMatch::Init (
    const char * fileName,
    const char * treeName,
    const char * branchVarName )

```

Setter version of [Init\(\)](#)

Same parameters as Regular ctor.

See also

[TRefMatch\(const char* fileName, const char* treeName, const char* branchVarName \)](#)

6.7.3.3 Prob2Sig()

```

Double_t TRefMatch::Prob2Sig (
    Double_t prob )

```

Convert probability to significance.

Convert a probability on (0,1) to its corresponding Gaussian significance.

Parameters

<i>prob</i>	– decimal probability on (0,1)
-------------	--------------------------------

Returns

sig – decimal significance on (0,inf)

See also

[Sig2Prob\(\)](#)

6.7.3.4 SetReferenceFileDir() [1/2]

```
void TRefMatch::SetReferenceFileDir (
    const char * refFileDirName )
```

Set directory (by name) to search for reference files.

Parameters

<i>refFileDirName</i>	– name of directory to search
-----------------------	-------------------------------

6.7.3.5 SetReferenceFileDir() [2/2]

```
void TRefMatch::SetReferenceFileDir (
    TSystemDirectory * refFileDir )
```

Set directory (by pointer) to search for reference files.

Parameters

<i>refFileDir</i>	– directory to search
-------------------	-----------------------

6.7.3.6 SetReferenceFilePattern() [1/2]

```
void TRefMatch::SetReferenceFilePattern (
    const char * pattern )
```

Set search pattern (by string) for reference filenames.

Parameters

<i>pattern</i>	– string to use for regex for filenames
----------------	---

6.7.3.7 SetReferenceFilePattern() [2/2]

```
void TRefMatch::SetReferenceFilePattern (
    TRegexp patternRE )
```

Set search pattern (by TRegexp object) for reference filenames.

Parameters

<i>patternRE</i>	– regex for filenames
------------------	-----------------------

6.7.3.8 Sig2Prob()

```
Double_t TRefMatch::Sig2Prob (
    Double_t sig )
```

Convert significance to probability.

Convert a significance to its corresponding Gaussian probability.

Parameters

<i>sig</i>	– decimal significance on (0,inf)
------------	-----------------------------------

Returns

prob – decimal probability on (0,1)

See also

[Prob2Sig\(\)](#)

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- TRefMatch.cxx

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